

Nguyen Dang Trac

AI Automation Engineer

tracng417@gmail.com

0384-170-128

github.com/trac41799

linkedin.com/in/trac-nguyen

Summary

M.S. in Electrical Engineering with 3+ years of experience spanning AI research, embedded software (Bosch), and production-grade AI-native product engineering at Edge8 AI. Built multiple multi-tenant SaaS products — from-scratch multi-agent systems with custom LLM orchestration, RAG pipelines, streaming AI UX, and clinical risk engines with MCP-based agent layers. First-author IEEE Transactions publication; Best Paper Award recipient.

Experience

AI Automation Engineer — Edge8 AI

Jan 2025 – Present

- Built multiple multi-tenant SaaS products and internal platforms (AI-native, content, analytics, regulated health-tech) — all on Supabase (PostgreSQL) and Vercel.
- Designed from-scratch production multi-agent AI systems (router/planner/executor) with custom multi-model LLM client (OpenRouter — Anthropic, Gemini, OpenAI), model fallback, cost-optimized routing, and streaming SSE via sse-starlette.
- Built full-stack platforms with Next.js (App Router), FastAPI, Supabase/PostgreSQL, and Redis; delivered real-time AI UX (streaming reasoning, TTS/STT), 7-locale i18n, Stripe billing, and RAG pipelines with hybrid search (pgvector, FAISS, BM25).
- Enforced engineering quality: unit/integration/E2E tests (Vitest, Pytest, Playwright), CI/CD, tenant-isolation test suites, and LLM evaluation harnesses with persona-based eval suites.

Research Assistant — University of Ulsan, South Korea

Sep 2022 – Aug 2024

- Led deep learning research on error correction coding (ECC); published first-author in *IEEE Trans. Cogn. Commun. Netw.* and won Best Paper Award at ICGHIT 2024.
- Processed multi-modal datasets and designed Transformer, CNN, RNN architectures in PyTorch/TensorFlow/Keras for wireless communication systems.

Embedded Software Engineer — Bosch Global Software Vietnam

Jun 2021 – Aug 2022

- Developed and validated automotive embedded software for Engine Control Units (ECU) across full SDLC; programmed in C, Python, XML with CAN/FlexRay protocols.
- Delivered features under Agile Scrum and V-Model; maintained CI/CD pipelines on a Jenkins-based platform.

Recent Projects

AI Learning-Management Platform

FastAPI, pgvector, Supabase, Vercel

- Built a from-scratch multi-agent AI engine (router→planner→executor) with a hand-rolled multi-model LLM client — streaming, multimodal input, prompt caching, model fallback, and cost-optimized routing via OpenRouter (Anthropic, Gemini, OpenAI).
- Implemented AI grading with human-in-the-loop review: rubric-based scoring (0–100), score-gated final assignments, and professor review states; custom RAG pipeline (multi-strategy chunking, pgvector hybrid search); data sync across 3 integrated systems (20+ tools). White-label variant with JWKS auth and org-level data isolation. (1,300+ commits)

Multi-Tenant Travelling Assistant Platform

Next.js, FastAPI, LlamaIndex, FAISS, Supabase

- Full-stack consumer + business platform with AI chat, streaming reasoning, voice I/O (Google Cloud STT + TTS, edge-tts), and a hybrid routing layer (rule-based multilingual fast-path + LLM router). LlamaIndex-powered agentic backend with conversational + research agents.
- Built RAG enrichment workspace: FAISS vector + BM25 hybrid retrieval, ~1K-token chunks, Gemini + VoyageAI embeddings, and live research-progress streaming. 7-locale i18n (EN, FR, JA, KO, RU, VI, ZH) with Redux Toolkit / RTK Query data layer. Companion native mobile clients scaffolded. (1,500+ commits)
- Benchmarked retrieval strategies on the production corpus: single-pass hybrid retrieval achieves 0.97 facet recall at both k=5 and k=10 — 3.8–4.0× faster than multi-query and agentic alternatives, which were shown to reduce recall via fixed-budget rank dilution. Designed a deterministic routing rule lifting knowledge-query activation from 0.64 to 1.00 at zero false-activation rate; cross-modal retrieval with curated-text captions reaches 0.955 top-1 accuracy (vs 0.136 for model-vision captions).

Occupational Health & Longevity Analytics Suite

Next.js 16, TypeScript, Vercel AI SDK, MCP, Supabase

- Built a multi-domain clinical risk engine (TypeScript) scoring 6 health domains — cardiovascular, metabolic, musculoskeletal, neurodegenerative, oncological, and biological age — with per-domain regression tests and snapshot suites for numerical stability.
- Designed AI assistant layer with DB-stored persona agents (clinician, nutrition, PT, supplement-advisor) exposed via Vercel AI SDK + MCP tool-calling. Built the Atlas MSK chatbot: intent classifier (Gemini Flash) + Claude synthesis, cache-first + MSSQL fan-out query layer, and pgvector RAG running in parallel. Delivered an LLM evaluation harness with per-persona eval suites, PII redaction, and RLS security hardening. (200+ commits)

Claude Code Agent Orchestration Plugin

Python, TypeScript, MCP, FastMCP, Claude Code

- Built a Claude Code plugin and desktop extension — an installable agent-template system with clean lifecycle management hooks (pre/post-execution, handoff-document contracts, validation gates) and a plugin repository that installs directly into the Claude Code runtime (github.com/talentedgeai/infiniteleverage-plugin). Adopted by clients across the US, Australia, and Vietnam for rapid POC and branding website delivery.
- Authored a universal orchestrator (assess→plan→distribute→collect→validate→report) that spawns parallel sub-agent swarms across Claude Code, OpenCode, and Cursor; designed 8 role-based agent templates (PM, Developer, QA, DevOps, Designer, Writer, Publisher, Marketer) with standalone skill files and MCP servers exposing curated CRUD and agentic tools to LLM clients.

Skills

Languages: Python, TypeScript, C, SQL (PostgreSQL/PLpgSQL), XML

AI & ML: PyTorch, TensorFlow, Keras, LlamaIndex, Vercel AI SDK, MCP (FastMCP), pgvector, FAISS, BM25 — RAG, Hybrid Search, Streaming LLM, Multi-Agent Orchestration, LLM Evaluation

Frameworks & Tools: Next.js, React, FastAPI, Redux Toolkit, TanStack Query, Tailwind, Supabase, Redis, Stripe, Sentry, OpenRouter, Google Cloud (STT/TTS/Translate)

Testing & DevOps: Playwright, Vitest, Pytest, CI/CD (Jenkins, GitHub Actions), Git, Agile/Scrum, V-Model

Agent-Assisted Development: Claude Code, OpenCode, Devin (Windsurf) — workflow automation, code generation, review, agent template authoring

Education

M.S. Electrical Engineering — University of Ulsan, South Korea *Sep 2022 – Aug 2024* GPA 4.21/4.5 Thesis: Enhanced Syndrome-based Reliability Decoding for Error Correction Codes Transformer

B.S. Electrical, Electronics & Telecommunication — Ton Duc Thang University, Vietnam *Sep 2017 – May 2022* Thesis: Application of MLP Model in Federated Learning

Publications

- D.-T. Nguyen & S. Kim, “U-shaped Error Correction Code Transformers,” *IEEE Trans. Cogn. Commun. Netw.*, Oct. 2024. *(First Author)*
- T. A. Khoa *et al.*, “Safety Is Our Friend,” *IEEE Trans. Comput. Social Syst.*, vol. 11, Apr. 2024. *(Second Author)*
- N. D. Trac & K. Sunghwan, “DRF-ECCT,” *Proc. ICGHIT*, Feb. 2024, pp. 18–23. *(Best Paper Award)*

Awards & Languages

Best Paper Award — ICGHIT 2024
(Native), English (Fluent, IELTS 7.0)

AF-1 Full Scholarship — Univ. of Ulsan

Languages: Vietnamese